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Air conditioner and method for operating an air conditioner based on user activity

Abstract

An air conditioner and a method for operating an air conditioner are provided. The air conditioner may include a case having a space formed therein and an open bottom; a panel disposed under the case and having an inlet and a plurality of outlets disposed around a circumference of the inlet; a fan disposed in the case and configured to generate an airflow from the inlet to the plurality of outlets; an airflow direction controller disposed at the plurality of outlets and configured to control an airflow direction of air flowing through the plurality of outlets in a plurality of directions; a camera disposed at one side of the panel and directed toward an indoor space; and a controller configured to control the airflow direction controller based on images corresponding to the indoor space and captured by the camera. The controller may be configured to determine an activity level of an occupant for each of a plurality of areas of the indoor space based on a plurality of images corresponding to the indoor space; determine a direction of air discharged from the plurality of outlets based on the determined activity level; and control the airflow direction controller according to the direction of the air which is determined for the plurality of outlets.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
10371399	12/2018	Rodriguez	N/A	F24F 3/0442
2005/0279118	12/2004	Son	62/428	F24F 1/0011
2017/0039729	12/2016	Wang	N/A	G06T 7/254
2018/0209681	12/2017	Song	N/A	F24F 11/77
2019/0309978	12/2018	Song	N/A	F24F 11/58
2019/0353386	12/2018	Lee	N/A	F24F 11/79
2019/0368773	12/2018	Lee	N/A	F24F 13/10
2021/0088244	12/2020	Kim	N/A	F24F 11/79
2021/0215373	12/2020	Han	N/A	F24F 11/79
2021/0247095	12/2020	Lee	N/A	F24F 11/79
2022/0083782	12/2021	Qian	N/A	G06V 20/41
2022/0201900	12/2021	Ma	N/A	H05K 7/20736
2022/0230474	12/2021	Hasedzic	N/A	G06V 20/60
2023/0066057	12/2022	Fan	N/A	F24F 11/62
2023/0375207	12/2022	Klein	N/A	F24F 11/65

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
06-323599	12/1993	JP	N/A
2010-121797	12/2009	JP	N/A
2011-174648	12/2010	JP	N/A

2011-185591	12/2010	JP	N/A
2013-253717	12/2012	JP	N/A
2015-137836	12/2014	JP	N/A
2016-200311	12/2015	JP	N/A
2018-514743	12/2017	JP	N/A
2021-048797	12/2020	JP	N/A
10-2012-0025955	12/2011	KR	N/A
10-1333661	12/2012	KR	N/A

OTHER PUBLICATIONS

Japanese Office Action issued in Application No. 2022-115689 dated Jul. 11, 2023. cited by applicant

European Search Report dated Dec. 6, 2022 issued in Application No. 22186432.5. cited by applicant

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S)

(1) This application claims priority under 35 U.S.C. § 119 to Korean Application No. 10-2021-0096539, filed in Korea on Jul. 22, 2021, whose entire disclosure is hereby incorporated by reference.

BACKGROUND

1. Field

(2) An air conditioner and a method for operating an air conditioner are disclosed herein.

2. Background

(3) An air conditioner is a device for supplying cooled/heated air into a room to control an indoor temperature and to purify indoor air, thereby providing a comfortable indoor environment.

Generally, the air conditioner includes an indoor unit including a heat exchanger and installed in an indoor space, and an outdoor unit including a compressor, and a heat exchanger, for example, and supplying refrigerant to the indoor unit.

(4) The air conditioner operates in a cooling mode or in a heating mode depending on a flow of refrigerant. During the cooling operation, high-temperature and high-pressure liquid refrigerant is supplied from the compressor of the outdoor unit by passing through the heat exchanger of the outdoor unit into the indoor unit, and the refrigerant is expanded and vaporized in the heat exchanger of the indoor unit such that an ambient air temperature decreases, and as an indoor unit fan is rotated, cooled air is discharged into the indoor space. During the heating operation, high-temperature and high-pressure liquid refrigerant is supplied from the compressor of the outdoor unit into the indoor unit, and air is heated by energy which is released as the high-temperature and high-pressure liquid refrigerant is liquefied in the heat exchanger of the indoor unit, and the heated air is discharged to the indoor space by the rotation of the indoor unit fan.

(5) As for an existing air conditioner, a user may adjust a direction of air by operating an airflow direction control means of the air conditioner directly with a hand or using a remote control device. However, it can be very uncomfortable for the user to manually operate the airflow direction control means, and if the user is located away from the air conditioner or is unable to locate a

position of the remote control device or loses the remote control device, the user may not manually operate the airflow direction control means in some cases. Accordingly, various studies have been conducted to provide an optimal airflow for users even without requiring them to manually operate the airflow direction control means.

(6) Korean Laid-open Patent Publication No. 10-2012-0025955 (related art 1), which is hereby incorporated by reference, discloses a method of sensing an occupant's movement using an infrared sensor, determining an activity level of the occupant according to a type of sound detected by a microphone, and controlling an indoor temperature based on the occupant's activity level. In addition, Korean Patent No. 10-1333661 (related art 2), which is hereby incorporated by reference, discloses a method of determining a moving distance of an occupant present in a space using an image captured by a camera provided in an indoor unit, and controlling indoor temperature based on an activity level corresponding to the moving distance.

(7) However, when the microphone is used, sounds may be generated inside of and outside of the space at the same time, such that various sounds may be input to the microphone. Accordingly, the method disclosed in related art 1 has a problem in that the accuracy in determining various sounds simultaneously input to the microphone is significantly reduced, such that the method may not accurately determine an activity level of an occupant.

(8) Further, an activity level of an occupant doing intense exercise at a fixed position is generally greater than an activity level when the occupant simply moves. However, the method disclosed in related art 2 may determine that the activity level when the occupant simply moves is greater because a moving distance is longer, such that it is difficult to provide an air-conditioned environment that is optimized for the actual activity level of the occupant.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Embodiments will be described in detail with reference to the following drawings in which like reference numerals refer to like elements, and wherein:

(2) FIG. 1 is a perspective view of an air conditioner according to an embodiment;

(3) FIG. 2 is a cross-sectional view of one side of the air conditioner according to the embodiment of FIG. 1;

(4) FIG. 3 is a diagram explaining an outlet and an airflow direction control unit of the air conditioner according to the embodiment of FIG. 1;

(5) FIGS. 4A to 4C are diagrams explaining an arrangement at each position of the airflow direction control unit according to the embodiment of FIG. 1;

(6) FIGS. 5A and 5B are diagrams explaining an airflow range according to an arrangement of an airflow direction control unit in a heating mode and in a cooling mode of an air conditioner according to an embodiment;

(7) FIG. 6 is a block diagram of an air conditioner according to an embodiment;

(8) FIG. 7 is a schematic diagram of an outdoor unit and an indoor unit according to an embodiment;

(9) FIGS. 8A to 8C are diagrams referred to in the description of the operation of an airflow direction control unit provided in an air conditioner according to an embodiment;

(10) FIGS. 9 and 10 are flowcharts of a method for operating an air conditioner according to an embodiment;

(11) FIGS. 11 to 13 are diagrams referred to in the description of the operation of an air conditioner according to an activity level of an occupant;

(12) FIG. 14 is a cross-sectional view of one side of the air conditioner according to another embodiment;

- (13) FIGS. **15A** to **15C** are diagrams explaining an arrangement at each position of the airflow direction control unit according to the embodiment of FIG. **14**;
- (14) FIG. **16** is a cross-sectional view of one side of an air conditioner according to still another embodiment;
- (15) FIGS. **17A** to **17C** are diagrams explaining an arrangement at each position of the airflow direction control unit according to the embodiment of FIG. **16**;
- (16) FIG. **18** is a cross-sectional view of one side of an air conditioner according to still another embodiment; and
- (17) FIGS. **19A** to **19C** are diagrams explaining an airflow at each rotational speed of an airflow direction control unit according to the embodiment of FIG. **18**.

DETAILED DESCRIPTION

- (18) Hereinafter, embodiments will be described in detail with reference to the accompanying drawings. In order to clearly and briefly describe embodiments, components that are irrelevant to the description will be omitted in the drawings. The same reference numerals are used throughout the drawings to designate the same or similar components, and a redundant description thereof will be omitted.
- (19) Terms “module” and “unit” for elements used in the following description are given simply in view of the ease of the description, and do not carry any important meaning or role. Therefore, the terms “module” and “unit” may be used interchangeably.
- (20) It should be understood that the terms, “comprise”, “include”, “have”, etc. when used in this specification, specify the presence of stated features, integers, steps, operations, elements, components, or combinations thereof but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, or combinations thereof.
- (21) It will be understood that, although the terms first, second, etc., may be used herein to describe various elements, these elements should not be limited by these terms. These terms are only used to distinguish one element from another.
- (22) Hereinafter, embodiments will be described with reference to the accompanying drawings.
- (23) Referring to FIGS. **1** to **4C**, an air conditioner **100** according to an embodiment will be described hereinafter. The air conditioner **100** may be, for example, a ceiling mounted air conditioner **100**.
- (24) Referring to FIG. **1**, the air conditioner **100** may include an inlet **122** which opens downwardly, and an outlet **124** which is disposed around the inlet **122** and opens downwardly. Referring to FIG. **2**, the air conditioner **100** may include a case **110** having a space formed therein and an open bottom; a panel **120** disposed under the case **110** and having the inlet **122** and the outlet **124** formed therein; a fan **112** disposed in the case **110**; a fan motor **114** that rotates the fan **112**; a heat exchanger **116** that performs heat exchange with air blown by the fan **112**; and an airflow direction control unit (airflow direction controller) **130** disposed at the outlet **124** and configured to control an airflow direction of flowing air.
- (25) Referring to FIG. **1**, the outlet **124** may include a plurality of outlets **124a**, **124b**, **124c**, and **124d** formed in the panel **120** and spaced apart from each other in different directions around the circumference of the inlet **123**. The outlet **124** may include first outlet **124a**, second outlet **124b**, third outlet **124c**, and fourth outlet **124d**. The first outlet **124a**, the second outlet **124b**, the third outlet **124c**, and the fourth outlet **124d** may be sequentially adjacent to each other and may be perpendicular to each other with respect to the inlet **122**. The airflow direction control unit **130** may include first airflow direction control unit **130a**, second airflow direction control unit **130b**, third airflow direction control unit **130c**, and fourth airflow direction control unit **130d** disposed at the first outlet **124a**, the second outlet **124b**, the third outlet **124c**, and the fourth outlet **124d**, respectively.
- (26) The configuration and description of one outlet **124** and airflow direction control unit **130** disposed therein, which will be described hereinafter, may also be applied to the other outlets and

airflow direction control units disposed therein, and thus are designated by the common reference numerals.

(27) Referring to FIG. 3, the airflow direction control unit **130** may include a first vane **140** coupled to two links **160** and **170**, and a second vane **150** coupled to one of the links coupled to the first vane **140** and having one side rotatably coupled to the panel **120**. The first vane **140** and the second vane **150**, which are disposed in each of the first outlet **124a**, the second outlet **124b**, the third outlet **124c**, and the fourth outlet **124d**, may be disposed at different positions.

(28) The first vane **140** may cover the outlet **124** or may be disposed under the outlet **124**. The first vane **140** may be formed longer than the second vane **150**.

(29) Referring to FIG. 3, while the fan **112** is in a stopped state in which the fan is not in operation, the first vane **140** may be disposed below the second vane **150**. The first vane **140** may include a first vane plate **142** that guides an airflow direction of flowing air, and a first link plate **144** that protrudes forwardly from both ends of left and right or lateral sides of the first vane plate **142** and coupled to the plurality of links **160** and **170**.

(30) The second vane **150** may include a second vane plate **152**, a second link plate **154** that protrudes forwardly from both ends of the left and right sides of the second vane plate **152** and coupled to a third link **180**, and a connector **156** disposed in a left-right or lateral direction of the second vane plate **152** and coupled to the panel **120**. The second vane plate **152** may have a curved shape. However, in another example, the second vane plate **152** may have a flat shape.

(31) Referring to FIG. 3, the airflow direction control unit **130** may include the first link **160** rotatably coupled to the panel **120** and the first vane **140**, and the second link **170** spaced apart from the first link **160** and rotatably coupled to the panel **120** and the first vane **140**. Referring to FIG. 3, the airflow direction control unit **130** may include the third link **180** rotatably coupled to one end of the first link **160** and the second vane **150**.

(32) The first link **160** may be rotatably coupled to the first vane **140** and the second vane **150**. The first link **160** may be coupled to a vane motor (not shown) to be rotated thereby. The first link **160** may include a panel connector **162** rotatably coupled to the panel **120**; a first link bar **164** that extends toward the first vane **140** and having an end rotatably coupled to the first vane **140**; and a second link bar **166** that extends from the panel connector **162** toward the second vane **150** and having an end rotatably coupled to the second vane **150**.

(33) Referring to FIG. 4A, a length **164L** of the first link bar **164** may be formed longer than a length **166L** of the second link bar **166**. The length **164L** of the first link bar **164** may be formed shorter than a length **170L** of the second link **170**. The length **164L** of the first link bar **164** may be formed longer than a length **180L** of the third link **180**. The first link **160** may be disposed closer to the inlet **122** than the second link **170**.

(34) An arrangement of the first vane **140** may be changed by the first link **160** and the second link **170**. As the arrangement of the first vane **140** is changed by the first link **160** and the second link **170**, the first vane **140** may be spaced apart downwardly from the outlet **124**. The first vane **140** may be elevated downwardly, and then an inclination thereof may be changed in a direction perpendicular to a ground. A first end **151a** of the second vane **150** may be moved downwardly, and then the first end **151a** may be moved inwardly and outwardly according to the arrangement of the third link **180**.

(35) Referring to FIG. 3, in the following description of the first vane **140** and the second vane **150**, ends of each of the first vane **140** and the second vane **150**, which are disposed far away from the inlet **122**, are defined as first ends **141a** and **151a**, respectively, and ends of each of the first vane **140** and the second vane **150**, which are disposed adjacent to the inlet **122**, are defined as second ends **141b** and **151b**, respectively. The second vane **150** may be rotatably coupled to the panel **120** at a position inwardly from the first vane **140**. A direction adjacent to the inlet **122** may be set as an inward direction, and a direction further away from the inlet **122** may be set as an outward direction.

(36) Referring to FIGS. 4A to 4C, the airflow direction control unit **130** may adjust an airflow direction of air discharged through the outlet **124** according to the arrangement. Referring to FIG. 4A, the airflow direction control unit **130** may be disposed in a first position **P1** in which the air discharged through the outlet **124** flows in a direction horizontal to the ground. Referring to FIG. 4A, when the airflow direction control unit **130** is disposed in the first position **P1**, the first vane **140** may be disposed approximately horizontal to the ground. Referring to FIG. 4A, when the airflow direction control unit **130** is disposed in the first position **P1**, the first vane **140** may form a first inclination angle θ_1 of 30 degrees or less with respect to a virtual horizontal line parallel to the ground. The first inclination angle θ_1 is formed between the first vane **140** and the virtual horizontal line which is parallel to the ground, and may vary depending on the arrangement of the first vane **140**.

(37) Referring to FIG. 4A, when the airflow direction control unit **130** is disposed in the first position **P1**, the second end **141b** of the first vane **140** may be disposed adjacent to the first end **151a** of the second vane **150**. The second end **141b** of the first vane **140** may be directed toward the first end **151a** of the second vane **150**.

(38) Referring to FIG. 4A, when the airflow direction control unit **130** is disposed in the first position **P1**, the first inclination angle θ_1 , formed between the first vane **140** and the virtual horizontal line, may be smaller than a second inclination angle θ_2 between a virtual line, connecting the first end **141a** and the second end **151b** of the second vane **150**, and the virtual horizontal line (or a “second inclination angle between the second vane and the virtual horizontal line”). The second inclination angle θ_2 is formed between the virtual line, connecting the first end **151a** and the second end **151b** of the second vane **150**, and the virtual horizontal line, and may vary depending on the arrangement of the second vane **150**.

(39) Accordingly, air flowing downwardly through the outlet **124** may sequentially flow along the second vane **150** and the first vane **140**. Referring to FIG. 4A, when the airflow direction control unit **130** is disposed in the first position **P1**, the air discharged through the outlet **124** may flow in a direction horizontal to the ground.

(40) Referring to FIG. 4B, the first vane **140** and the second vane **150** may be disposed in a second position **P2** in which the air discharged through the outlet **124** flows in a direction perpendicular to the ground. Referring to FIG. 4B, when the airflow direction control unit **130** is disposed in the second position **P2**, the first vane **140** may be disposed approximately perpendicular to the ground. Referring to FIG. 4B, when the airflow direction control unit **130** is disposed in the second position **P2**, the first vane **140** may form a first inclination angle θ_1 of 60 degrees or more with respect to the virtual horizontal line parallel to the ground.

(41) Referring to FIG. 4B, when the airflow direction control unit **130** is disposed in the second position **P2**, the second end **141b** of the first vane **140** may be spaced apart from the first end **151a** of the second vane **150**. Referring to FIG. 4B, when the airflow direction control unit **130** is disposed in the second position **P2**, the second end **141b** of the first vane **140** may be disposed above the first end **151a** of the second vane **150**.

(42) Referring to FIG. 4B, when the airflow direction control unit **130** is disposed in the second position **P2**, the second end **141b** of the first vane **140** may be directed above the second end **151b** of the second vane **150**. Referring to FIG. 4B, when the airflow direction control unit **130** is disposed in the second position **P2**, the first vane **140** and the second vane **150** may be disposed approximately parallel to each other.

(43) Referring to FIG. 4B, when the airflow direction control unit **130** is disposed in the second position **P2**, the first inclination angle θ_1 , formed between the first vane **140** and the virtual horizontal line, may be similar to the second inclination angle θ_2 formed between the second vane **150** and the virtual horizontal line. Referring to FIG. 4B, when the airflow direction control unit **130** is disposed in the second position **P2**, the air discharged through the outlet **124** may flow in a direction perpendicular to the ground.

(44) Referring to FIG. 4C, the first vane **140** and the second vane **150** may be disposed in a third position P3 in which the air discharged through the outlet **124** flows in a direction diagonal to the ground. When the airflow direction control unit **130** is disposed in the third position P3, the air discharged by passing through the first vane **140** and the second vane **150** may form a diagonal airflow which is directed downwardly below a horizontal airflow in the first position P1 and is directed upwardly above a vertical airflow in the second position P2.

(45) Referring to FIG. 4C, when the airflow direction control unit **130** is disposed in the third position P3, the first vane **140** may be disposed at an inclination angle between the first vane **140** when the airflow direction control unit **130** is disposed in the first position P1, and the first vane **140** when the airflow direction control unit **130** is disposed in the second position P2. Referring to FIG. 4C, when the airflow direction control unit **130** is disposed in the third position P3, the first vane **140** may form a first inclination angle **61**, ranging from 30 degrees to 60 degrees, with respect to the virtual horizontal line parallel to the ground.

(46) When the airflow direction control unit **130** is disposed in the third position P3, a separated distance between the second end **141b** of the first vane **140** and the first end **151a** of the second vane **150** may be longer than a separated distance between the second end **141b** of the first vane **140** and the first end **151a** of the second vane **150** when the airflow direction control unit **130** is disposed in the first position. When the airflow direction control unit **130** is disposed in the third position P3, a separated distance between the second end **141b** of the first vane **140** and the first end **151a** of the second vane **150** may be shorter than a separated distance between the second end **141b** of the first vane **140** and the first end **151a** of the second vane **150** when the airflow direction control unit **130** is disposed in the second position P2.

(47) When the airflow direction control unit **130** is disposed in the first position P1, a horizontal airflow may be generated so that the air discharged through the outlet may flow in a direction horizontal to the ground. When the airflow direction control unit **130** is disposed in the second position P2, a vertical airflow may be generated so that the air discharged through the outlet may flow in a direction perpendicular to the ground. When the airflow direction control unit **130** is disposed in the third position P3, a diagonal airflow may be generated so that the air discharged through the outlet may flow in a direction between an indirect airflow and the vertical airflow. The horizontal airflow and the indirect airflow may be used interchangeably, or the vertical airflow and direct airflow may be used interchangeably.

(48) Referring to FIGS. 5A and 5B, the air discharged through the outlet **124** may be divided into three directions by the airflow direction control unit **130**. When the airflow direction control unit **130** is disposed in the first position P1, the air discharged through the outlet **124** may flow in a lateral direction. When the airflow direction control unit **130** is disposed in the second position P2, the air discharged through the outlet **124** may flow in a downward direction. When the airflow direction control unit **130** is disposed in the third position P3, the air discharged through the outlet **124** may flow in an intermediate direction between the lateral direction and the downward direction.

(49) In a vertical airflow range according to the first position P1, the second position P2, and the third position P3 of the airflow direction control unit **130**, an airflow when the airflow direction control unit **130** is in the third position P3 may be generated below an airflow generated when the airflow direction control unit **130** is in the first position P1 and may be generated above an airflow generated when the airflow direction control unit **130** is in the second position P2.

(50) Under cooling and heating conditions, the vertical airflow range may vary according to the first position P1, the second position P2, and the third position P3 of the airflow direction control unit **130**. Referring to FIGS. 5A and 5B, when the airflow direction control unit **130** is disposed in the first position P1, the air discharged through the outlet **124** may flow in a first direction D1. The first direction D1 may refer to a direction in which an angle formed between a main flow direction of the air discharged through the outlet **124** and the ground is in a range of zero degrees to 30

degrees. In the first direction **D1**, the angle may be formed in a same range under both the heating and cooling conditions.

(51) Referring to FIGS. 5A and 5B, when the airflow direction control unit **130** is disposed in the second position **P2**, the air discharged through the outlet **124** may flow in a second direction **D2**. In the second direction **D2**, the angle may be formed in different ranges under the heating and cooling conditions. Referring to FIG. 5A, under the heating condition, the second direction **D2** may refer to a direction in which the angle formed between the main flow direction of the air discharged through the outlet **124** and the ground is in a range of 60 degrees to 90 degrees. Referring to FIG. 5B, under the cooling condition, the second direction **D2** may refer to a direction in which the angle formed between the main flow direction of the air discharged through the outlet **124** and the ground is in a range of 45 degrees to 90 degrees.

(52) Referring to FIGS. 5A and 5B, when the airflow direction control unit **130** is disposed in the third position **P3**, the air discharged through the outlet **124** may flow in a third direction **D3**. In the third direction **D3**, the angle may be formed in different ranges under the heating and cooling conditions. Referring to FIG. 5A, under the heating condition, the third direction **D3** may refer to a direction in which the angle formed between the main flow direction of the air discharged through the outlet **124** and the ground is in a range of 30 degrees to 60 degrees. Referring to FIG. 5B, under the cooling condition, the third direction **D3** may refer to a direction in which the angle formed between the main flow direction of the air discharged through the outlet **124** and the ground is in a range of 45 degrees to 60 degrees.

(53) The ranges of the angle in the first direction **D1**, the second direction **D2**, and the third direction **D3** are merely examples, and may vary according to a space in which the air conditioner is disposed or a structure of the air conditioner. Further, while FIGS. 5A and 5B illustrate the airflow is divided into three sections, the airflow may also be subdivided into four to six sections.

(54) FIG. 6 is a block diagram of an air conditioner according to an embodiment. FIG. 7 is a schematic diagram of an outdoor unit and an indoor unit according to an embodiment.

(55) Referring to FIGS. 6 and 7, the air conditioner **100** may include a communication unit **610**, a sensor unit **620**, a memory **600**, a fan driver **650** that drives a fan **651**, a compressor driver **640** that drives compressor **102**, and/or a controller **670**. The communication unit **610** may include at least one communication module. For example, the communication unit **610** may be provided for each of an outdoor unit **710** and an indoor unit **720**, and the outdoor unit **710** and the indoor unit **720** may transmit and receive data with each other.

(56) A communication method of the outdoor unit **710** and the indoor unit **720** may include, for example, not only a wired communication technology, such as communication using power lines, serial communication, for example, RS-485 serial communication, and communication using a refrigerant pipe, but also a wireless communication technology, such as Wi-Fi, Bluetooth, Beacon, and Zigbee, for example.

(57) The communication unit **610** may transmit and receive data to and from outside of the air conditioner **100**. For example, the communication unit **610** may access a server connected to an external network to transmit and receive data.

(58) The sensor unit **620** may include at least one sensor and may transmit data on a value, detected by the sensor, to the controller **670**. The sensor unit **620** may include a heat exchanger temperature sensor (not shown). For example, the heat exchanger temperature sensor may be disposed in an indoor heat exchanger **108** to detect a temperature of the indoor heat exchanger **108**.

(59) The sensor unit **620** may include a pipe temperature sensor (not shown) that detects a temperature of a refrigerant flowing through each pipe. The sensor unit **620** may include a pressure sensor (not shown) that detects a pressure of gaseous refrigerant flowing through each pipe. The sensor unit **620** may include an indoor temperature sensor (not shown) that detects an indoor temperature and/or an outdoor temperature sensor (not shown) that detects an outdoor temperature. The sensor unit **620** may include an indoor humidity sensor (not shown) that detects an indoor

humidity and/or an outdoor humidity sensor (not shown) that detects an outdoor humidity.

(60) The sensor unit **620** may include an image sensor (not shown) that converts light into an electrical signal. The image sensor may include a plurality of photodiodes corresponding to a plurality of pixels. The image sensor may include a charge-coupled device (CCD) image sensor or a complementary metal-oxide-semiconductor (CMOS) image sensor; however, embodiments are not limited thereto.

(61) The air conditioner **100** may further include a lens through which light that emanates from a subject passes, and a digital signal processor that configures and processes an image based on a signal output from the image sensor, for example. The digital signal processor may be implemented as at least a portion of the processor **670** or as a separate processor operating independently from the processor **670**. For example, in a case in which the digital signal processor is implemented as a separate processor, the image processed by the digital signal processor may be stored in the memory as it is or after being further processed by the controller **670**.

(62) Airflow direction control unit **630** may adjust an airflow direction of air discharged from the air conditioner **100**. The airflow direction control unit **630** may include at least one vane disposed adjacent to an outlet through which air is discharged, a vane motor that drives the vane, and an airflow direction control fan, for example. For example, according to one embodiment, the airflow direction control unit **630**, for example, airflow direction control unit **130** of FIG. 2, may include the first vane **140** coupled to two links **160** and **170**, and the second vane **150** coupled to one of the links coupled to the first vane **140** and having one side rotatably coupled to the panel **120**.

(63) The compressor driver **640** may drive the compressor **102**. The compressor driver **640** may include a rectifier (not shown) that rectifies AC power into DC power and outputs the rectified power; a DC terminal capacitor (not shown) that stores a ripple voltage from the rectifier; an inverter (not shown) including a plurality of switching elements to convert the smoothed DC power into 3-phase AC power having a predetermined frequency and output the power; and/or a compressor motor **102b** that drives the compressor **102** according to the three-phase AC power.

(64) The fan driver **650** may drive the fan **651** provided in the air conditioner **100**. For example, the fan **651** may include an outdoor fan **105a** and/or an indoor fan **112**.

(65) The fan driver **650** may include the rectifier (not shown) that rectifies AC power into DC power and outputs the rectified power; the DC terminal capacitor (not shown) that stores a ripple voltage from the rectifier; the inverter (not shown) including a plurality of switching devices to convert the smoothed DC power into 3-phase AC power having a predetermined frequency and output the power; and/or at least one motor that drives the fan **651** according to the 3-phase AC power. In addition, the fan driver **650** may include separate components that drives each of the outdoor fan **105a** and the indoor fan **112**. For example, the air conditioner **100** may include a first fan driver that drives the outdoor fan **105a** and a second fan driver that drives the indoor fan **112**.

(66) The memory **660** may store data on a reference value associated with the operation of each component provided in the air conditioner **100**. The memory **660** may store programs that process and control each signal in the controller **670**, and may store processed data as well as data to be processed. For example, the memory **660** may store application programs designed for performing various operations that may be processed by the controller **670**, and in response to a request of the controller **670**, the memory **660** may selectively provide some of the stored application programs. The memory **660** may include at least one of a volatile memory, for example, DRAM, SRAM, or SDRAM, for example, and a non-volatile memory, for example, flash memory, hard disk memory (HDD), and solid-state drive (SSD), for example.

(67) The controller **670** may include at least one processor and may control an overall operation of the air conditioner **100** using the processor included therein. The processor may be a general processor, such as a central processing unit (CPU), for example. Obviously, the processor may be a dedicated device, such as an ASIC, or other hardware-based processor, for example.

(68) The controller **670** may control the overall operation of the air conditioner **100**. The controller

670 may be connected to the respective components provided in the air conditioner **100** and may transmit and/or receive signals with the respective components to control the overall operation of the respective components.

(69) The controller **670** may control the operation of the fan driver **650** to change the RPM of the fan **651**. For example, by changing a frequency of three-phase AC power output to the outdoor fan motor **105b** under the control of the controller **670**, the fan driver **650** may change the RPM of the outdoor fan **105a**. For example, by changing a frequency of the three-phase AC power output to the indoor fan motor **114** under the control of the controller **670**, the fan driver **650** may change the RPM of the indoor fan **112**.

(70) The controller **670** may control the operation of the compressor driver **640** to change an operating frequency of the compressor **102**. For example, by changing the frequency of the three-phase AC power output to the compressor motor **102b** under the control of the controller **670**, the compressor driver **640** may change the operating frequency of the compressor **102**.

(71) The controller **670** may obtain data associated with the respective components provided in the air conditioner **100**. In this case, by considering a computational load, the controller **670** may obtain data associated with the respective components provided in the air conditioner **100** at predetermined time intervals during a predetermined period. The data associated with the respective components provided in the air conditioner **100** may include, for example, the operating frequency of the compressor **102**, the temperature of refrigerant flowing into the compressor **102**, the temperature of refrigerant discharged from the compressor **102**, the pressure of refrigerant flowing into the compressor **102**, the pressure of refrigerant discharged from the compressor **102**, a pipe temperature at an inlet side of the indoor unit **720**, a pipe temperature at an outlet side of the indoor unit **720**, and an opening degree of an Electric Expansion Valve (EEV), for example.

(72) The controller **670** may perform various operations based on the obtained data and may control the overall operation of the respective components included in the air conditioner **100** based on operation results. The controller **670** may process the image acquired by the image sensor. For example, the controller **670** may remove noise from the image or may perform signal processing on the image, such as gamma correction, color filter array interpolation, color matrix, color correction, and color enhancement, for example.

(73) The controller **670** may detect an object included in the image using at least one method. For example, the controller **670** may extract characteristic points from the image using a method, such as scale invariant feature transform (SIFT), histogram of oriented gradient (HOG), for example, and may detect the object included in the image based on the extracted characteristic points. In this case, the controller **670** may detect the object included in the image by determining a boundary of the object using an algorithm, such as support vector machine (SVM), and AdaBoost, for example.

(74) Based on the plurality of processed images, the controller **670** may detect a motion of the object included in the plurality of images. For example, using a dense optical flow method, the controller **670** may calculate a motion vector for a plurality of pixels constituting the object detected in the image, and may calculate the motion of the object based on the calculated motion vector. In this embodiment, the calculation is performed using the dense optical flow method; however, embodiments are not limited thereto, and a sparse optical flow method for calculating a motion vector of some characteristic pixels may also be used.

(75) The controller **670** may determine an activity level of the object detected in the image. For example, the controller **670** may determine, as an activity level of the object detected in the image, a value obtained by dividing a sum of magnitudes of motion vectors for pixels included in the object by the number of pixels included in the object.

(76) The controller **670** may be provided not only in the outdoor unit **710** and the indoor unit **720**, but also in a central controller that controls operation of the outdoor unit **710** and/or the indoor unit **720**. The outdoor unit **710** may include a compressor **102b** that compresses a refrigerant; compressor motor **102b** that drives the compressor **102b**; outdoor heat exchanger **104** that radiates

heat of the compressed refrigerant; the outdoor fan **105a** disposed on one side of the outdoor heat exchanger **104** and accelerating heat radiation from the refrigerant; motor **105b** that rotates the outdoor fan **105a**; expansion valve **106** that expands condensed refrigerant; cooling/heating switch valve **107** that changes a passage of the compressed refrigerant; and accumulator **103** that temporarily stores gaseous refrigerant to remove moisture and foreign matter from the refrigerant and supplies the refrigerant at constant pressure to the compressor.

(77) The indoor unit **720** may include indoor heat exchanger **108** disposed indoors and performing cooling/heating operations, and indoor fan **112** disposed on one side of the indoor heat exchanger **108** and accelerating heat radiation from the refrigerant; and a motor **114** for rotating the indoor fan **112**.

(78) FIGS. **8A** to **8C** are diagrams referred to in the description of the operation of an airflow direction control unit included in an air conditioner according to an embodiment. In the following description, directions of the air conditioner **100** may be defined based on an orthogonal coordinates system illustrated in FIGS. **8A** to **8C**. In the orthogonal coordinates system, the x-axis direction may be defined as a left and right or lateral direction of the air conditioner **100**. In this case, with respect to an origin point, a right or a first direction may be oriented in the direction of the positive x-axis, and a left or second direction may be oriented in the direction of the negative x-axis. The y-axis direction may be defined as a forward and backward direction of the air conditioner **100**. In this case, with respect to the origin point, a forward direction may be oriented in the direction of the positive y-axis, and a backward direction may be oriented in the direction of the negative y-axis. The z-axis direction may be defined as an upward and downward or vertical direction. With respect to the origin point, an upward direction may be oriented in the direction of the positive z-axis, and a downward direction may be oriented in the direction of the negative z-axis.

(79) Referring to FIG. **8A**, the air conditioner **100** may obtain an image corresponding to a predetermined indoor space **800** using camera **190**. The camera **190** may be implemented as a lens, an image sensor, and a digital signal processor, for example. For example, light emanating from the indoor space **800** may pass through the lens to be incident on the image sensor, and the image sensor may convert the light into an electrical signal. In this case, the signal output from the image sensor may be transmitted to the controller **670** as it is or after being processed by the digital signal processor **670**. In this embodiment, one camera **190** is provided in the air conditioner **100**; however, embodiments are not limited thereto, and two or more cameras **190** may be provided in the air conditioner **100**.

(80) Based on the signal transmitted from the image sensor and/or the digital signal processor, the controller **670** may obtain the image corresponding to the indoor space **800**. By processing the image corresponding to the indoor space **800**, the controller **670** may perform determination on the indoor space **800**.

(81) In the case in which the air conditioner **100** is mounted on a ceiling, the camera **190** may be directed downward from the air conditioner **100**. For example, the camera **190** may be disposed on one side of the panel **120** disposed under the case **110** of the air conditioner **100**.

(82) Referring to FIGS. **8B** and **8C**, the indoor space **800** may be divided into a plurality of areas. The plurality of areas may correspond to at least one of the plurality of outlets **124** of the air conditioner **100**.

(83) A first area **801** of the indoor space **800** may correspond to the first outlet **124** of the air conditioner **100**. A second area **802** of the indoor space **800** may correspond to the second outlet **124b** of the air conditioner **100**. A third area **803** of the indoor space **800** may correspond to the third outlet **124c** of the air conditioner **100**. A fourth area **804** of the indoor space **800** may correspond to the fourth outlet **124d** of the air conditioner **100**. A fifth area **805**, which is located to the left or laterally of the second area **802** of the indoor space **800**, may correspond to the second outlet **124b** of the air conditioner. A sixth area **806**, which is located to the right or laterally of the

fourth area **804**, may correspond to the fourth outlet **124d** of the air conditioner **100**. A seventh area **807**, surrounded by the first area **801** to the fourth area **804** of the indoor space **800**, may correspond to the first outlet **124a** to the fourth outlet **124d** of the air conditioner **100**.

(84) In one embodiment, the indoor space **800** may further include an area located in front of the first area **801** of the indoor space **800** and/or an area located behind the third area **803** of the indoor space. In this case, the area located in front of the first area **801** may correspond to the first outlet **124a** of the air conditioner **100**, and the area located behind the third area **803** may correspond to the third outlet **124c** of the air conditioner.

(85) By controlling operation of the airflow direction control unit **630**, the controller **670** may control airflow directions of air flowing toward the indoor space **800** for the respective corresponding areas. For example, by controlling the vane motor of the airflow direction control unit **630**, the controller **670** may generate an airflow of air flowing toward the first area **801** as a vertical airflow perpendicular to the ground, and an airflow of air flowing toward the second area **802** as a horizontal airflow horizontal to the ground. In this case, an inclination angle formed by the vanes **140** and **150** disposed at the first outlet **124a** may be different from an inclination angle formed by the vanes **140** and **150** disposed at the second outlet **124b**.

(86) FIGS. **9** and **10** are flowcharts of a method for operating an air conditioner according to an embodiment. Referring to FIG. **9**, the air conditioner **100** may acquire an image corresponding to a predetermined indoor space using the camera **190** in operation **S910**. For example, by processing a signal output from the image sensor included in the camera **190**, the air conditioner **100** may acquire the image corresponding to the indoor space.

(87) In operation **S920**, the air conditioner **100** may process the image corresponding to the indoor space and captured by the camera **190**. For example, the air conditioner **100** may remove noise from the image corresponding to the indoor space, or may perform signal processing on the image, such as color correction, or color enhancement, for example. For example, the air conditioner **100** may detect an object included in the image corresponding to the indoor space, using a support vector machine (SVM) algorithm, for example.

(88) In operation **S930**, the air conditioner **100** may determine whether a predetermined number or more of images are acquired. The predetermined number may refer to a minimum number of images for calculating the motion of an object present in an indoor space.

(89) If the number of obtained images is less than the predetermined number, the air conditioner **100** may repeatedly perform an operation of obtaining images corresponding to the indoor space using the camera **190** and processing the obtained images. For example, the air conditioner **100** may repeatedly acquire images corresponding to the indoor space using the camera **190** during a predetermined period of time, and may process the images acquired during the predetermined period of time and may store the processed images in the memory **660**.

(90) In operation **S940**, upon acquiring the predetermined number or more of the images, the air conditioner **100** may calculate an activity level of an occupant present in the indoor space based on the images. Referring to **1110** of FIG. **11**, the air conditioner **100** may acquire the predetermined number or more of images. The predetermined number or more of images may be processed by the controller **670** and may be stored in the memory **660**. The air conditioner **100** may detect the respective objects included in the predetermined number or more of images.

(91) Referring to **1120** of FIG. **11**, the air conditioner **100** may detect motion of the objects included in the predetermined number or more of the images. For example, the air conditioner **10** may calculate motion vectors for pixels of the objects detected in the predetermined number or more of images using a dense optical flow method, and may calculate the motion of the objects based on the calculated motion vectors.

(92) Referring to **1130** of FIG. **11**, the air conditioner **100** may detect an activity level of objects **131** to **1134** detected in the predetermined number or more of the images. For example, the air conditioner **10** may determine the activity level of a first object **1131** using a value obtained by

dividing a sum of magnitudes of the motion vectors for pixels included in the first object **1131** by the number of the pixels included in the first object **1131**. In this case, even when the occupant moves in place, the magnitudes of the motion vectors for the plurality of pixels of the object are accumulated according to the movement of the occupant, such that the air conditioner **100** may accurately determine the activity level of the occupant.

(93) The air conditioner **100** may determine an activity level for each of a plurality of corresponding areas of the indoor space. For example, if an occupant stays in a specific area among the plurality of areas of the indoor space, an activity level of the occupant, which is determined by the air conditioner **100**, may be matched to the specific area in which the occupant is present. In the case in which the occupant moves in the indoor space, the location of an object included in the plurality of images may be changed in each image.

(94) Referring to FIGS. **12A** and **12B**, while an occupant **1200** moves from a fourth area **804** to a sixth area **806** in an indoor space **800**, the location of the occupant **1200** may be detected in at least one of the fourth area **804** or the sixth area **806**. For example, the occupant **1200** may be detected in the fourth area **804** in a first acquired image and may be detected in the sixth area **806** in a last acquired image, among a predetermined number or more of images.

(95) In addition, during a predetermined period T when the occupant **1200** moves from the fourth area **804** to the sixth area **806**, a magnitude **1210** of a motion vector calculated for the fourth area **804** and a magnitude **1215** of a motion vector calculated for the sixth area **806** may be separated at a time point $t_{sub.1}$. In this case, in one embodiment, the air conditioner **100** may determine an area, corresponding to the location where the object is detected, in the last acquired image among the predetermined number or more of the images to be an area corresponding to the activity level of the object. For example, the air conditioner **100** may determine, as the activity level of the occupant **1200**, a value obtained by dividing a sum of the magnitudes **1210** and **1215** of the motion vectors, calculated while the occupant **1200** moves from the fourth area **804** to the sixth area **806**, by the number of pixels constituting the occupant **1200** in a predetermined image. Further, the air conditioner **100** may match the determined activity level of the occupant **1200** to the sixth area **806**, in which the occupant **1200** is detected in the last acquired image.

(96) In another embodiment, the air conditioner **100** may determine the activity level of the object for each area, in which the object is detected, in the predetermined number or more of the images, and may match the activity level of the occupant, which is determined for a specific area, to the specific area.

(97) For example, using a value obtained by dividing a sum of the magnitudes **1210** of the motion vectors calculated for the fourth area **804** by the number of pixels constituting the occupant **1200** in a predetermined image, the air conditioner **100** may determine the activity level of the occupant **1200** in the fourth area **804**, and may match the determined activity level of the occupant **1200** to the fourth area **804**. In addition, using a value obtained by dividing a sum of the magnitudes **1215** of the motion vectors calculated for the sixth area **806** by the number of pixels constituting the occupant **1200** in a predetermined image, the air conditioner **100** may determine the activity level of the occupant **1200** in the sixth area **806**, and may match the determined activity level of the occupant **1200** to the sixth area **806**.

(98) The air conditioner **100** may determine a total activity level corresponding to all the plurality of areas of the indoor space. For example, upon detecting a plurality of objects in the indoor space, the air conditioner **100** may determine the total activity level using a value, obtained by dividing a sum of magnitudes of motion vectors for all the pixels constituting the plurality of objects by the number of pixels constituting the plurality of objects.

(99) The activity level determined by the air conditioner **100** may correspond to metabolic equivalent of task (MET). For example, the air conditioner **100** may calculate the MET, corresponding to the activity level, using a predetermined equation and may determine the activity level based on the calculated MET.

(100) In one embodiment, the air conditioner **100** may determine the activity level based on MET data similar to the following Table 1. For example, if the MET is less than 1.0, the air conditioner **100** may determine the activity level to be 1; if the MET is greater than or equal to 1.0 and less than 2.0, the air conditioner **100** may determine the activity level to be 2; if the MET is greater than or equal to 2.0 and less than 3.0, the air conditioner **100** may determine the activity level to be 3; if the MET is greater than or equal to 3.0 and less than 4.0, the air conditioner **100** may determine the activity level to be 4; and if the MET is greater than or equal to 4.0, the air conditioner **100** may determine the activity level to be 5.

(101) TABLE-US-00001 TABLE 1 Intensity Type MET Low Sleep 0.9 reading, watching television 1.0 desk work (writing, typing, etc.) 1.4 Mid playing an instrument 2.0 walking (2 km/h) 2.5 walking (3 km/h) 2.9 High cleaning 3.5 bicycling 4.0 jump rope 5.0

(102) Referring to FIG. 9, in operation **S950**, the air conditioner **100** may control the airflow direction of air, flowing toward the indoor space, based on the calculated activity level of the occupant present in the indoor space. According to the activity level corresponding to the respective areas of the indoor space, the air conditioner **100** may control each airflow direction of the air flowing toward the plurality of areas.

(103) For example, during a cooling operation, if the activity level corresponding to the first to third areas **801** to **803** is 1 or 2, and the activity level corresponding to the fourth area **804** is 4, the air conditioner **100** may determine an airflow direction of air flowing toward the first to third areas **801** to **803** to be an indirect airflow, and an airflow direction of air flowing toward the fourth area **804** to be a direct airflow. In this case, the air conditioner **100** may control the airflow direction control unit **630** so that an inclination angle formed by the vanes **140** and **150** disposed at the first to third outlets **124a** to **124c** may correspond to the indirect airflow, and an inclination angle formed by the vanes **140** and **150** disposed at the fourth outlet **124d** may correspond to the direct airflow.

(104) For example, during the cooling operation, if the activity level corresponding to the fifth area **805** is 3, the air conditioner **100** may determine an airflow direction of air flowing toward the fifth area **805** to be a diagonal airflow. In this case, the air conditioner **100** may control the airflow direction control unit **630** so that an inclination angle formed by the vanes **140** and **150** disposed at the second outlet **124b** may correspond to the diagonal airflow.

(105) For example, during the cooling operation, if the activity level corresponding to the sixth area **806** is 4, the air conditioner **100** may determine an airflow direction of air flowing toward the sixth area **806** to be a diagonal airflow and a vertical airflow. In this case, the air conditioner **100** may control the airflow direction control unit **630** so that an inclination angle formed by the vanes **140** and **150** disposed at the fourth outlet **124b** may be continuously changed between the diagonal airflow and the vertical airflow.

(106) For example, during the cooling operation, if the activity level corresponding to the seventh area **807** is 3, the air conditioner **100** may determine an airflow direction of air flowing toward the seventh area **807** to be a diagonal airflow. In this case, the air conditioner **100** may control the airflow direction control unit **630** so that inclination angles formed by the vanes **140** and **150** disposed at the first to fourth outlets **124a** to **124b** all correspond to the diagonal airflow.

(107) For example, during a heating operation, if the activity level corresponding to the first area **801** is 1 or 2, and the activity level corresponding to the third area **803** is 4, the air conditioner **100** may determine an airflow direction of air flowing toward the first area **801** to be the direct airflow, and an airflow direction of air flowing toward the third area **803** to be the indirect airflow. In this case, the air conditioner **100** may control the airflow direction control unit **630** so that an inclination angle formed by the vanes **140** and **150** disposed at the first outlet **124a** may correspond to the direct airflow, and an inclination angle formed by the vanes **140** and **150** disposed at the third outlet **124c** corresponds to the indirect airflow.

(108) Further, the air conditioner **100** may also adjust the airflow direction of air flowing toward

the indoor space based on an airflow predetermined according to user preference. For example, in the case where a user sets an indirect airflow as a preferred airflow during the cooling operation, if an activity level corresponding to the first area **801** is greater than or equal to 4 and an activity level corresponding to the second to fourth areas **802** to **804** is 1 or 2, the air conditioner **100** may determine an airflow direction of air, flowing toward the first area **801**, to be the indirect airflow and may determine an airflow direction of air, flowing toward the second to fourth areas **802** to **804**, to be the direct airflow as an airflow different from the user's preferred airflow. In this case, the air conditioner **100** may control the airflow direction control unit **630** so that the angle formed by the vanes **140** and **150** disposed at the first outlet **124a** may correspond to the indirect airflow, and the angle formed by the vanes **140** and **150** disposed at the second to fourth outlets **124b** to **124d** may correspond to the direct airflow.

(109) In operation **S960**, the air conditioner **100** may control the temperature of the indoor space based on the calculated activity level of the occupant present in the indoor space. The air conditioner **100** may control the temperature of the indoor space bases on a total activity level corresponding to all the plurality of areas of the indoor space, which will be described hereinafter with reference to FIG. **10**.

(110) Referring to FIG. **10**, the air conditioner **100** may determine whether a predetermined period of time has elapsed in operation **S1010**. The predetermined period of time may refer to a period during which the air conditioner **100** determines whether to change a target temperature in order to control the temperature of the indoor space. For example, the air conditioner **100** may control the temperature of the indoor space at predetermined five-minute intervals based on the calculated activity level of the occupant present in the indoor space.

(111) If the predetermined period of time has not elapsed, the air conditioner **100** may control the temperature of the indoor space according to a current set target temperature. If the predetermined period of time has elapsed, the air conditioner **100** may determine whether a cooling operation is performed in operation **S1020**.

(112) If the cooling operation is performed, the air conditioner **100** may determine whether a total activity level during the predetermined period of time is maintained at a level less than a predetermined level in operation **S1030**. For example, if the total activity level corresponding to all of the plurality of areas during a five-minute period is less than 3, the air conditioner **100** may determine that the total activity level during the predetermined period of time is maintained at a level less than the predetermined level.

(113) If the total activity level during the predetermined period of time is maintained at a level less than the predetermined level, the air conditioner **100** may set a temperature higher than the existing target temperature as a new target temperature in **S1040**. That is, if the total activity level during the predetermined period of time is maintained at a level less than the predetermined level, the air conditioner **100** may determine that an activity level of the occupant is sufficiently low, and may increase the target temperature to reduce power consumption.

(114) The existing target temperature may refer to a target temperature determined according to a desired temperature set by a user. For example, in the case where a desired temperature set by a user is 18° C., if cooling performance is at a first level, the air conditioner **100** may determine the existing target temperature to be 23° C., if cooling performance is at a second level, the air conditioner **100** may determine the existing target temperature to be 21° C., and if cooling performance is at a third level, the air conditioner **100** may determine the existing target temperature to be 18° C.

(115) Upon setting a higher temperature than the existing target temperature as a new target temperature, the air conditioner **100** may determine the new target temperature based on the determined total activity level during the predetermined period of time. For example, if an average of the total activity levels during the predetermined period of time is less than 2, the air conditioner **100** may determine a temperature higher than the existing target temperature by a first temperature,

for example, 2° C., as the new target temperature, and if an average of the total activity levels during the predetermined period of time is greater than or equal to 2, the air conditioner **100** may determine a temperature higher than the existing target temperature by a second temperature, for example, 1° C., as the new target temperature.

(116) If the new target temperature, determined based on the total activity level during the predetermined period of time, is greater than or equal to a maximum target temperature, for example, 28° C., during the cooling operation, the air conditioner **100** may set the maximum target temperature, for example, 28° C., as the new target temperature. Alternatively, if the new target temperature, determined based on the total activity level during the predetermined period of time, is greater than or equal to a maximum target temperature, for example, 28° C., during the cooling operation, the air conditioner **100** may set the existing target temperature as the new target temperature.

(117) The air conditioner **100** may control the indoor temperature based on the new target temperature in operation **S1050**. For example, if a current indoor temperature is lower than the new target temperature by a reference value, for example, 1° C., or more, the air conditioner **100** may decrease an operating frequency of the compressor **102**, and if the current indoor temperature is greater than or equal to the new target temperature by the reference value, for example, 1° C., or more, the air conditioner **100** may increase an operating frequency of the compressor **102**.

(118) During a heating operation, the air conditioner **100** may determine whether the total activity level during the predetermined period of time is maintained at a level greater than or equal to the predetermined level in operation **S1060**. For example, if an average of total activity levels corresponding to the plurality of areas during a five-minute period is greater than or equal to 3, the air conditioner **100** may determine that the total activity level during the predetermined period of time is maintained at a level greater than or equal to the predetermined level.

(119) If the total activity level during the predetermined period of time is maintained at a level greater than or equal to the predetermined level, the air conditioner **100** may set a temperature lower than the existing target temperature as a new target temperature in operation **S1070**. That is, if the total activity level is maintained at a level greater than or equal to the predetermined level during the predetermined period of time, the air conditioner **100** may determine that the activity level of an occupant is sufficiently high and may lower the target temperature to reduce power consumption.

(120) The existing target temperature may refer to a target temperature determined according to a desired temperature set by a user. For example, in the case where a desired temperature set by a user is 21° C., if heating performance is at a first level, the air conditioner **100** may determine the existing target temperature to be 21° C., if heating performance is at a second level, the air conditioner **100** may determine the existing target temperature to be 24° C., and if heating performance is at a third level, the air conditioner **100** may determine the existing target temperature to be 26° C.

(121) When setting a temperature lower than the existing target temperature as the new target temperature, the air conditioner **100** may determine the new target temperature based on the total activity level determined during the predetermined period of time. For example, if an average of the total activity levels during the predetermined period of time is less than 4, the air conditioner **100** may determine a temperature lower than the existing target temperature by a first temperature, for example, 1° C., as the new target temperature, and if an average of the total activity levels during the predetermined period of time is greater than or equal to 4, the air conditioner **100** may determine a temperature lower than the existing target temperature by a second temperature, for example, 2° C., as the new target temperature.

(122) If the new target temperature, determined based on the total activity level during the predetermined period of time, is lower than a minimum target temperature, for example, 18° C., during the heating operation, the air conditioner **100** may set the minimum target temperature, for

example, 18° C., as the new target temperature. Alternatively, if the new target temperature, determined based on the total activity level during the predetermined period of time, is lower than the minimum target temperature, for example, 18° C., during the heating operation, the air conditioner **100** may set the existing target temperature as the new target temperature.

(123) The air conditioner **100** may control the indoor temperature based on the new target temperature in operation **S1080**. For example, if a current indoor temperature is higher than the new target temperature by a reference value, for example, 1° C., or more, the air conditioner **100** may decrease an operating frequency of the compressor **102**, and if the current indoor temperature is lower than the new target temperature by the reference value, for example, 1° C., or more, the air conditioner **100** may increase an operating frequency of the compressor **102**.

(124) If the total activity level during the predetermined period of time is greater than or equal to the predetermined level during the cooling operation of the air conditioner **100**, or if the total activity level during the predetermined period of time is less than the predetermined level during the heating operation the air conditioner **100**, the air conditioner **100** may control the indoor temperature based on the existing target temperature in operation **S1090**. In this case, the predetermined level used as a reference for determining the total activity level during the cooling operation may be the same as or different from the predetermined level used as a reference for determining the total activity during the heating operation.

(125) Referring to FIG. **13**, the air conditioner **100** may perform the cooling operation based on an indoor temperature **1300**, a variable target temperature **1310**, and operating temperatures **1320** and **1325** set according to the existing target temperature.

(126) Upon initiating the cooling operation, the air conditioner **100** may set the existing target temperature according to the desired temperature set by a user. For example, if the desired temperature set by the user is 21° C., the air conditioner **100** may set the existing target temperature to 24° C. In this case, each of the operating temperatures **1320** and **1325** may be set to be different from the determined existing target temperature by a predetermined temperature value, for example, 1° C.

(127) The air conditioner **100** may perform the cooling operation so that the indoor temperature **1300** reaches the existing target temperature. In this case, the air conditioner **100** may increase a cooling capacity to a maximum level, so that at an initial stage after the cooling operation is initiated, the indoor temperature **1300** may rapidly reach the existing target temperature. For example, at the initial stage after initiating the cooling operation, the air conditioner **100** may increase the operating frequency of the compressor **102** to a predetermined maximum frequency.

(128) In the case in which the indoor temperature **1300** reaches a first operating temperature **1320**, the air conditioner **100** may lower the cooling capacity. For example, the air conditioner **100** may lower the operating frequency of the compressor **102** starting from a first time point $t_{sub.1}$ when the indoor temperature **1300** reaches the first operating temperature **1320**. In the case in which the indoor temperature **1300** reaches a second operating temperature **1325**, the air conditioner **100** may lower the operating frequency of the compressor to a predetermined minimum frequency.

(129) If a stabilization stage is satisfied in which an airflow and/or temperature is maintained at a predetermined level during a predetermined period $T_{sub.1}$ from the first time point $t_{sub.1}$ at which the indoor temperature **1300** reaches the first operating temperature **1320**, the air conditioner **100** may activate a power saving mode based on the total activity level of an occupant, and starting from a time point $t_{sub.2}$ at which the power saving mode is activated, the air conditioner **100** may perform the cooling operation based on the total activity level of the occupant.

(130) Starting from the time point $t_{sub.2}$ at which the power saving mode is activated, the air conditioner **100** may acquire images corresponding to the indoor space using the camera **190**. In addition, the air conditioner **100** may reset the target temperature at predetermined time intervals $T_{sub.2}$ according to the total activity level of the occupant calculated based on the acquired images.

(131) For example, if an average of the total activity levels of the occupant which are calculated during a period from the time point $t_{sub.2}$ to a third time point $t_{sub.3}$ is less than 2, the air conditioner **100** may determine a temperature higher than the existing target temperature **1310** by a first temperature, for example, 2° C., to be a new target temperature, and may control the indoor temperature **1300** based on the determined new target temperature.

(132) For example, if an average of the total activity levels of the occupant which are calculated during a period from a time point $t_{sub.4}$ to a fifth time point $t_{sub.5}$ is greater than or equal to 2, the air conditioner **100** may determine a temperature higher than the existing target temperature **1310** by a second temperature, for example, 1° C., to be the new target temperature, and may control the indoor temperature **1300** based on the determined new target temperature.

(133) Hereinafter, an air conditioner **200** according to another embodiment will be described with reference to FIGS. **14** to **15C**. The air conditioner **200** according to this embodiment is different from the air conditioner **100** according to the previous embodiment in terms of a configuration of the airflow direction control unit **230**. Accordingly, description of components other than the airflow direction control unit **230** may be replaced with the above description of the air conditioner **100** according to the previous embodiment.

(134) The airflow direction control unit **230** of the air conditioner **200** according to this embodiment may include one vane **240** disposed at each outlet **224**, and a vane motor (not shown) that drives the vane **240**. An arrangement of the vane **240** may be changed by operation of the vane motor.

(135) Referring to FIGS. **15A** to **15C**, by varying an inclination angle of the vane **240** disposed at the outlet **224**, the airflow direction control unit **230** may control an airflow direction of air flowing through the outlet **224**. The vane **240** may be disposed so as to close the outlet **224** or to adjust the airflow direction of air flowing through the outlet **224**.

(136) Referring to FIG. **15A**, the airflow direction control unit **230** may be disposed in a first position **P1** in which the vane **240** is disposed approximately parallel to a virtual horizontal line parallel to a ground. When the airflow direction control unit **230** is disposed in the first position **P1**, the vane **240** may form an inclination angle θ of 30 degrees or less with respect to a virtual horizontal line **HL** parallel to the ground. The inclination angle θ , which is formed between the vane **240** and the virtual horizontal line **HL**, may vary according to the arrangement of the vane **240**.

(137) Referring to FIG. **15B**, the airflow direction control unit **230** may be disposed in a second position **P2** in which the vane **240** is disposed approximately perpendicular to the horizontal line parallel to the ground. When the airflow direction control unit **230** is disposed in the second position **P2**, the vane **240** may form an inclination angle θ of 60 degrees or more with respect to the virtual horizontal line **HL**.

(138) Referring to FIG. **15C**, the airflow direction control unit **230** may be disposed in a third position **P3** in which the vane **240** forms an angle between the first position **P1** and the second position **P2**. When the airflow direction control unit **230** is disposed in the third position **P3**, the vane **240** may form an inclination angle θ , ranging from 30 degrees to 60 degrees, with respect to the virtual horizontal line **HL**.

(139) When the airflow direction control unit **230** is disposed in the first position **P1**, an indirect airflow is formed so that air, discharged through the outlet, may flow in a direction horizontal to the ground. When the airflow direction control unit **230** is disposed in the second position **P2**, a vertical airflow is formed so that air, discharged through the outlet, may flow in a direction perpendicular to the ground. When the airflow direction control unit **230** is disposed in the third position **P3**, a diagonal airflow may be generated so that the air discharged through the outlet may flow in a direction between the indirect airflow and the vertical airflow.

(140) As illustrated in FIGS. **5A** and **5B**, the air conditioner according to this embodiment may also cause the air discharged through the outlet **224** to flow in the first direction **D1**, the second

direction D2, and the third direction D3 according to the first position P1, the second position P2, and the third position P3 of the airflow direction control unit 230.

(141) Hereinafter, an air conditioner according to still another embodiment will be described with reference to FIGS. 16 to 17C. Air conditioner 300 according to this embodiment is different from the air conditioner 100 according to the embodiment of FIG. 1 in terms of a configuration and operating structure of the air conditioner and the airflow direction control unit, as well as in terms of a shape of an outlet and an arrangement of a vane. Accordingly, description of components other than the shape of the outlet and the airflow direction control unit may be replaced with the above description of the air conditioner according to the embodiment of FIG. 1.

(142) The air conditioner 300 according to this embodiment may have a plurality of outlets 324 formed on an outer circumference of an inlet 322. The inlet 322 may have a square shape, and the outlets 324 may be spaced apart outwardly from each side of the inlet 322. In addition, the inlet 322 may also have a circular shape. In this case, there may be a plurality of outlets 324 formed at positions spaced apart radially from an outer circumference of the circular inlet 322.

(143) The outlets 324 formed at the air conditioner 300 according to this embodiment may have an outer end 324b disposed above an inner end 324a. Further, a discharge passage 325 formed over the outlet 324 may have a structure extending outwardly from an upper side toward a lower side.

(144) The airflow direction control unit 330 of the air conditioner according to this embodiment may include a vane 340 disposed at a panel 320 and protruding to the outlet 324 with a variable protruding length; a vane motor (not shown) disposed in the panel 320 and driving the vane 340; and a vane gear 350 rotated by the vane motor and engaged with the vane 340 to move the arrangement of the vane 340. The vane 340 may have one end engaged with the vane gear 350 in a rack gear structure.

(145) The vane 340 may be disposed at the inner end 324a of the outlet 324. The vane 340 may protrude outwardly from the inner end 324a of the outlet 324. The protruding length of the vane 340 that protrudes into the outlet 324 may be changed by operation of the vane motor. The airflow direction control unit 330 may control an airflow direction of air, flowing through the outlet 324, according to the protruding length of the vane 340 that protrudes into the outlet 324.

(146) Referring to FIGS. 17A to 17C, by varying the protruding length of the vane 340 that protrudes into the outlet 324, the airflow direction control unit 330 may control the airflow direction of the air flowing through the outlet 324. Referring to FIG. 17A, the airflow direction control unit 330 may be disposed in a first position P1 in which the vane 340 protrudes into the outlet 324 to a maximum length. When the airflow direction control unit 330 is disposed in the first position P1, the vane 340 may protrude to a maximum allowable length. Accordingly, when the airflow direction control unit 330 is disposed in the first position P1, the air flowing through the outlet 335 may be guided in a direction horizontal to the ground. When the airflow direction control unit 330 is disposed in the first position P1, the vane 340 is arranged below outer end 324b of the outlet 324, such that the air discharged through the outlet 324 may flow along the vane 340 in the direction horizontal to the ground.

(147) Referring to FIG. 17B, the airflow direction control unit 330 may be disposed in a second position P2 in which the vane 230 does not protrude into the outlet 324. When the airflow direction control unit 330 is disposed in the second position P2, the vane 230 may be disposed so as not to be exposed to the outlet 324. Accordingly, when the airflow direction control unit 330 is disposed in the second position P2, the air flowing through the discharge passage 335 may be discharged through the outlet 324 in a direction approximately perpendicular to the ground. However, depending on a shape of the discharge passage 335, a portion of the air flowing through the outlet 324 may also flow diagonally to the ground.

(148) Referring to FIG. 17C, the airflow direction control unit 330 may be disposed in a third position P3 in which the vane 340 protrudes to a length shorter than the protruding length of the vane 340 when the airflow direction control unit 330 is disposed in the first position P1. When the

airflow direction control unit **330** is disposed in the third position **P3**, the vane **340** may protrude into a length longer than the protruding length of the vane **340** that protrudes to the outlet **324** when the airflow direction control unit **330** is disposed in the second position **P2**. When the airflow direction control unit **330** is disposed in the third position **P3**, the vane **340** may protrude to a length which is $\frac{1}{3}$ to $\frac{2}{3}$ of the protruding length of the vane **340** that protrudes to the outlet **324** when the airflow direction control unit **330** is disposed in the second position **P2**.

(149) When the airflow direction control unit **330** is disposed in the first position **P1**, an indirect airflow may be formed so that the air discharged through the outlet may flow in a direction horizontal to the ground. When the airflow direction control unit **330** is disposed in the second position **P2**, a vertical airflow may be formed so that the air discharged through the outlet may flow in a direction perpendicular to the ground. When the airflow direction control unit **330** is disposed in the third position **P3**, a diagonal airflow may be formed so that the air discharged through the outlet may flow in a direction between the indirect airflow and the vertical airflow.

(150) As illustrated in FIGS. 5A and 5B, the air conditioner according to this embodiment may also cause the air discharged through the outlet **324** to flow in the first direction **D1**, the second direction **D2**, and the third direction **D3** according to the first position **P1**, the second position **P2**, and the third position **P3** of the airflow direction control unit **330**.

(151) Hereinafter, an air conditioner according to yet another embodiment will be described with reference to FIGS. 18 to 19C. Air conditioner **400** according to this embodiment is different from the air conditioner **100** according to the embodiment of FIG. 1 in terms of a configuration of the air conditioner **100** and the airflow direction control unit according to the embodiment of FIG. 1. In the air conditioner according to this embodiment, an inlet **422** has a circular shape, and an outlet **424** is formed in an annular shape around the inlet **422**.

(152) An airflow direction control unit **430** of the air conditioner according to this embodiment may include an airflow direction control fan **440** disposed on one side of the outlet **424**. The airflow direction control unit **430** may include a motor **450** that rotates the airflow direction control fan **440**. The airflow direction control fan **440** may be disposed on one side in a direction where the inlet **422** is disposed and in a region where the outlet **424** is formed, and may adjust an airflow direction of air discharged through the outlet **424**. The airflow direction control fan **440** may be disposed on one side of the outlet **424** to adjust an airflow direction of air discharged through the outlet **424**. A plurality of airflow direction control fans **440** may be formed in an annular circumferential direction where the outlet **424** is formed.

(153) The airflow direction control fan **440** may draw in ambient air around the outlet **424** to change pressure, thereby adjusting the airflow direction of the air flowing to the outlet **424**. The airflow direction control fan **440** may control an amount of the drawn ambient air around the outlet **424**.

(154) By regulating a rotational speed of the airflow direction control fan **440** or by stopping the airflow direction control fan **440**, the airflow direction control unit **430** may control the airflow direction of the air discharged through the outlet **424**. When the airflow direction control fan **440** stops, the air flowing to the outlet **424** is affected by a shape of the discharge passage **425** and an opening direction of the outlet **424**. Accordingly, when the airflow direction control fan **440** stops, the air flowing through the outlet **424** may be discharged in a direction perpendicular to the ground.

(155) However, when the airflow direction control fan **440** is operated, a portion of the air discharged through the outlet **242** is affected by the airflow direction control fan **440**, such that the air discharged through the outlet **424** may flow diagonally in a direction horizontal to the ground. In this case, the air flowing to the outlet **424** may be controlled based on the amount of air drawn by the airflow direction control fan **440**. When the rotational speed of the airflow direction control fan **440** increases, the amount of air drawn by the airflow direction control fan **440** increases, thereby causing the air to flow in a direction parallel to the ground.

(156) Referring to FIGS. 19A to 19C, the airflow direction control unit **430** may control the airflow

direction of the air discharged through the outlet **424** by controlling the operation or rotational speed of the airflow direction control fan **440**. Referring to FIG. **19A**, the airflow direction control unit **430** may rotate at a first setting speed for rotating the airflow direction control fan **440** at a maximum rotational speed. By rotating at the first setting speed, the airflow direction control unit **430** may generate an indirect airflow for causing the air discharged through the outlet to flow in a direction horizontal to the ground.

(157) Referring to FIG. **19B**, the airflow direction control unit **430** may rotate at a second setting speed for rotating the airflow direction control fan **440** at a minimum rotational speed or stopping the airflow direction control fan **440**. The second setting speed corresponds to a speed including a “zero” speed. Accordingly, the second setting speed of the airflow direction control unit **430** may include a state in which the airflow direction control fan **440** stops. By rotating at the second setting speed, the airflow direction control unit **430** may generate a vertical airflow for causing the air discharged through the outlet to flow in a direction perpendicular to the ground.

(158) Referring to FIG. **19C**, the airflow direction control unit **430** may rotate at a third setting speed for rotating the airflow direction control fan **440** at a rotational speed ranging between the first setting speed and the second setting speed. By rotating at the third setting speed, the airflow direction control unit **430** may generate a diagonal airflow for causing the air discharged through the outlet to flow in a direction between the indirect airflow and the vertical airflow.

(159) By rotating at the first rotational speed which is the maximum speed, the airflow direction control unit **430** may generate an indirect airflow for causing the air discharged through the outlet to flow in a direction horizontal to the ground. By rotating at the second rotational speed for rotation at the minimum speed or stopping, the airflow direction control unit **330** may generate a vertical airflow for causing the air discharged through the outlet to flow in a direction perpendicular to the ground. By rotating at the third rotational speed corresponding to a speed between the first rotational speed and the second rotational speed, the airflow direction control unit **430** may generate a diagonal airflow for causing the air discharged through the outlet in a direction between the indirect airflow and the vertical airflow.

(160) As illustrated in FIGS. **5A** and **5B**, the air conditioner according to this embodiment may also cause the air discharged through the outlet **324** to flow in the first direction **D1**, the second direction **D2**, and the third direction **D3** according to the first position **P1**, the second position **P2**, and the third position **P3** of the airflow direction control unit **330**.

(161) The method for operating an air conditioner according to the embodiments of FIGS. **9** and **10** may be applied to the air conditioner according to embodiments disclosed herein.

(162) Embodiments disclosed herein solve the above and other problems.

(163) Embodiments disclosed herein provide an air conditioner capable of accurately determining an occupant's activity level, and a method for operating an air conditioner. Embodiments disclosed herein further provide an air conditioner capable of providing an optimal airflow for an occupant based on a position and an activity level of the occupant, and a method for operating an air conditioner. Embodiments disclosed herein furthermore provide an air conditioner capable of controlling a temperature of an indoor space appropriately based on an activity level of an occupant, and a method for operating an air conditioner.

(164) Advantages are not limited to the aforementioned advantages and other advantages not described herein will be clearly understood by those skilled in the art from the following description.

(165) Embodiments disclosed herein provide an air conditioner that may include a plurality of outlets which open downwardly; a plurality of airflow direction control units (airflow direction controllers) disposed at the respective outlets; and a camera disposed to be directed toward an indoor space. A controller may determine an activity level of an occupant for each of a plurality of areas of the indoor space based on a plurality of images corresponding to the indoor space; determine a direction of air discharged from the respective outlets based on the determined activity

level; and control the airflow direction control unit according to the direction of the air which is determined for the respective outlets.

(166) According to an embodiment, among the plurality of areas, if an activity level for a first area is greater than a predetermined level and an activity level for a second area is less than the predetermined level, the controller may control the airflow direction control unit so that a direction of air discharged from a first outlet corresponding to the first area is different from a direction of air discharged from a second outlet corresponding to the second area in the upward and downward direction. During a cooling operation, the controller may control the airflow direction control unit so that the direction of the air discharged from the first outlet may be vertically below the direction of the air discharged from the second outlet, and during a heating operation, the controller may control the airflow direction control unit so that the direction of the air discharged from the first outlet is vertically above the direction of the air discharged from the second outlet. Among the plurality of areas, if an activity level corresponding to a third area satisfies a criterion associated with the predetermined level, the controller may control the airflow direction control unit so that a direction of air discharged from a third outlet corresponding to the third area may change continuously between the direction of the air discharged from the first outlet and the direction of the air discharged from the second outlet.

(167) According to an embodiment, the air conditioner may further include a temperature sensor configured to sense a temperature of the indoor space. The controller may set a target temperature based on a result of comparison between a predetermined reference and a sum of activity levels of the occupant which are determined for each of the plurality of areas, and may control the temperature of the indoor space based on a difference between a current temperature of the indoor space, which is sensed by the temperature sensor, and the target temperature.

(168) The controller may calculate a motion vector for a plurality of pixels constituting the occupant and included in the plurality of images using a dense optical flow method, and may determine the activity level of the occupant based on the calculated motion vector. The controller may determine, as the activity level of the occupant, a value obtained by dividing a sum of magnitudes of motion vectors for the plurality of pixels, which are calculated for the plurality of images, by a number of the pixels.

(169) According to an embodiment, the airflow direction control unit may include a plurality of vanes disposed at the respective outlets. By adjusting an arrangement of each of the vanes, the controller may change the direction of the air discharged from the respective outlets.

(170) The airflow direction control unit may include a plurality of airflow direction control fans, each disposed on one side of the respective outlets, wherein by regulating a rotational speed of the plurality of airflow direction control fans, the controller may change the direction of the air discharged from the respective outlets.

(171) Embodiments disclosed herein provide a method for operating an air conditioner that may include determining an activity level of an occupant for each of a plurality of areas of an indoor space based on a plurality of images corresponding to the indoor space and captured by a camera of the air conditioner; determining a direction of air discharged from the respective outlets based on the determined activity level; and according to the direction of the air which is determined for the respective outlets, controlling an airflow direction control unit configured to control an airflow direction of air flowing through the respective outlets in an upward and downward direction. The controlling of the airflow direction control unit may include a first operation of controlling the airflow direction control unit in which, among the plurality of areas, if an activity level for a first area is greater than a predetermined level and an activity level for a second area is less than the predetermined level, the airflow direction control unit is controlled so that a direction of air discharged from a first outlet corresponding to the first area is different from a direction of air discharged from a second outlet corresponding to the second area in the upward and downward direction.

(172) The first operation of controlling the airflow direction control unit may include a second operation of controlling the airflow direction control unit in which, during a cooling operation, the airflow direction control unit is controlled so that the direction of the air discharged from the first outlet is vertically below the direction of the air discharged from the second outlet; and a third operation of controlling the airflow direction control unit in which, during a heating operation, the airflow direction control unit is controlled so that the direction of the air discharged from the first outlet is vertically above the direction of the air discharged from the second outlet. The controlling of the airflow direction control unit may include, among the plurality of areas, if an activity level corresponding to a third area satisfies a criterion associated with the predetermined level, controlling the airflow direction control unit so that a direction of air discharged from a third outlet corresponding to the third area may change continuously between the direction of the air discharged from the first outlet and the direction of the air discharged from the second outlet.

(173) According to an embodiment, the method for operating an air conditioner may further include setting a target temperature based on a result of comparison between a predetermined reference and a sum of activity levels of the occupant which are determined for each of the plurality of areas, and controlling temperature of an indoor space based on a difference between a current temperature of the indoor space, which is sensed by a temperature sensor that senses the temperature of the indoor space, and the target temperature. The determining of the activity level of the occupant may include using a dense optical flow method, calculating a motion vector for a plurality of pixels constituting the occupant and included in the plurality of images, and determining the activity level of the occupant based on the calculated motion vector. The determining of the activity level of the occupant based on the calculated motion vector may include determining, as the activity level of the occupant, a value obtained by dividing a sum of magnitudes of motion vectors for the plurality of pixels, which are calculated for the plurality of images, by a number of the pixels.

(174) According to an embodiment, the airflow direction control unit may include a plurality of vanes disposed at the respective outlets. The controlling of the airflow direction control unit may include changing the direction of the air discharged from the respective outlets by adjusting an arrangement of each of the vanes.

(175) According to an embodiment, the airflow direction control unit may include a plurality of airflow direction control fans, each disposed on one side of the respective outlets. The controlling of the airflow direction control unit may include changing a direction of the air discharged from the respective outlets by regulating a rotational speed of the plurality of airflow direction control fans. A distance between the third area and the air conditioner may be longer than at least one of a distance between the first area and the air conditioner and a distance between the second area and the air conditioner.

(176) Embodiments disclosed herein have one or more of the following advantages.

(177) First, based on a result of processing pixels included in images of an indoor space, an activity level of an occupant may be determined accurately. Second, by determining a position and activity level of an occupant for each of a plurality of areas of an indoor space, a direction of air supplied to respective areas may be adjusted to various airflow directions, thereby providing an optimal airflow for the occupant. Third, by comprehensively determining activity levels of all the occupants present in the indoor space, a target temperature optimized for the occupants may be set, such that the temperature of the indoor space may be controlled appropriately.

(178) Embodiments are not limited to the aforesaid, and other effects not described herein will be clearly understood by those skilled in the art from the description of the appended claims.

(179) The attached drawings are provided to help easy understanding of the embodiments, not limiting the scope and spirit. Thus, it is to be understood that embodiments cover all modifications, equivalents, and/or alternatives falling within the scope and spirit.

(180) Similarly, while operations are depicted in the drawings in a particular order, this should not be understood as requiring that such operations be performed in the particular order shown or in

sequential order, or that all illustrated operations be performed, to achieve desirable results. In certain circumstances, multitasking and parallel processing are advantageous.

(181) While embodiments have been particularly shown and described with reference to embodiments thereof, it will be understood by those skilled in the art that embodiments are not limited to those exemplary embodiments and various changes in form and details may be made therein without departing from the scope and spirit as defined by the appended claims and should not be individually understood from the technical spirit or prospect.

(182) It will be understood that when an element or layer is referred to as being “on” another element or layer, the element or layer can be directly on another element or layer or intervening elements or layers. In contrast, when an element is referred to as being “directly on” another element or layer, there are no intervening elements or layers present. As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items.

(183) It will be understood that, although the terms first, second, third, etc., may be used herein to describe various elements, components, regions, layers and/or sections, these elements, components, regions, layers and/or sections should not be limited by these terms. These terms are only used to distinguish one element, component, region, layer or section from another region, layer or section. Thus, a first element, component, region, layer or section could be termed a second element, component, region, layer or section without departing from the teachings of the present invention.

(184) Spatially relative terms, such as “lower”, “upper” and the like, may be used herein for ease of description to describe the relationship of one element or feature to another element(s) or feature(s) as illustrated in the figures. It will be understood that the spatially relative terms are intended to encompass different orientations of the device in use or operation, in addition to the orientation depicted in the figures. For example, if the device in the figures is turned over, elements described as “lower” relative to other elements or features would then be oriented “upper” relative to the other elements or features. Thus, the exemplary term “lower” can encompass both an orientation of above and below. The device may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein interpreted accordingly.

(185) The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of the invention. As used herein, the singular forms “a”, “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “comprises” and/or “comprising,” when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

(186) Embodiments are described herein with reference to cross-section illustrations that are schematic illustrations of idealized embodiments (and intermediate structures). As such, variations from the shapes of the illustrations as a result, for example, of manufacturing techniques and/or tolerances, are to be expected. Thus, embodiments should not be construed as limited to the particular shapes of regions illustrated herein but are to include deviations in shapes that result, for example, from manufacturing.

(187) Unless otherwise defined, all terms (including technical and scientific terms) used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this invention belongs. It will be further understood that terms, such as those defined in commonly used dictionaries, should be interpreted as having a meaning that is consistent with their meaning in the context of the relevant art and will not be interpreted in an idealized or overly formal sense unless expressly so defined herein.

(188) Any reference in this specification to “one embodiment,” “an embodiment,” “example embodiment,” etc., means that a particular feature, structure, or characteristic described in connection with the embodiment is included in at least one embodiment. The appearances of such

phrases in various places in the specification are not necessarily all referring to the same embodiment. Further, when a particular feature, structure, or characteristic is described in connection with any embodiment, it is submitted that it is within the purview of one skilled in the art to effect such feature, structure, or characteristic in connection with other ones of the embodiments.

(189) Although embodiments have been described with reference to a number of illustrative embodiments thereof, it should be understood that numerous other modifications and embodiments can be devised by those skilled in the art that will fall within the spirit and scope of the principles of this disclosure. More particularly, various variations and modifications are possible in the component parts and/or arrangements of the subject combination arrangement within the scope of the disclosure, the drawings and the appended claims. In addition to variations and modifications in the component parts and/or arrangements, alternative uses will also be apparent to those skilled in the art.

Claims

1. An air conditioner, comprising: a case having a space formed therein and an open bottom; a panel disposed under the case and having an inlet and a plurality of outlets disposed around a circumference of the inlet; a fan disposed in the case and configured to generate an airflow from the inlet to the plurality of outlets; an airflow direction controller disposed at the plurality of outlets and configured to control an airflow direction of air flowing through the plurality of outlets in a plurality of directions; a camera disposed at one side of the panel and directed toward an indoor space; and a controller configured to control the airflow direction controller based on images corresponding to the indoor space and captured by the camera, wherein the controller is configured to: calculate a motion vector for a plurality of pixels constituting an occupant and included in the plurality of images; determine an activity level of the occupant for each of a plurality of areas of the indoor space; determine a direction of air discharged from the plurality of outlets based on the determined activity level; and control the airflow direction controller according to the direction of the air which is determined for the plurality of outlets, wherein the activity level of the occupant is determined based on a value obtained by dividing a sum of magnitudes of motion vectors for the plurality of pixels, which are calculated for the plurality of images, by a number of the pixels.
2. The air conditioner of claim 1, wherein among the plurality of areas, based on an activity level for a first area being greater than a predetermined level and an activity level for a second area being less than the predetermined level, the controller is configured to control the airflow direction controller so that a direction of air discharged from a first outlet corresponding to the first area is different from a direction of air discharged from a second outlet corresponding to the second area.
3. The air conditioner of claim 2, wherein the controller is configured to: during a cooling operation, control the airflow direction controller so that the direction of the air discharged from the first outlet is vertically below the direction of the air discharged from the second outlet; and during a heating operation, control the airflow direction controller so that the direction of the air discharged from the first outlet is vertically above the direction of the air discharged from the second outlet.
4. The air conditioner of claim 2, wherein among the plurality of areas, based on an activity level corresponding to a third area satisfying a criterion associated with the predetermined level, the controller is configured to control the airflow direction controller so that a direction of air discharged from a third outlet corresponding to the third area changes continuously between the direction of the air discharged from the first outlet and the direction of the air discharged from the second outlet.
5. The air conditioner of claim 4, wherein a distance between the third area and the air conditioner is longer than at least one of a distance between the first area and the air conditioner or a distance

between the second area and the air conditioner.

6. The air conditioner of claim 1, further comprising a temperature sensor configured to sense a temperature of the indoor space, wherein the controller is configured to: set a target temperature based on a result of a comparison between a predetermined reference and a sum of activity levels of the occupant which are determined for each of the plurality of areas; and control the temperature of the indoor space based on a difference between a current temperature of the indoor space, which is sensed by the temperature sensor, and the target temperature.

7. The air conditioner of claim 1, wherein the motion vector is calculated using an optical flow method.

8. The air conditioner of claim 1, wherein the airflow direction controller comprises a plurality of vanes disposed at the plurality of outlets, respectively, and wherein the controller is configured to change the direction of the air discharged from the plurality of outlets by adjusting an arrangement of each of the plurality of vanes.

9. The air conditioner of claim 1, wherein the airflow direction controller comprises a plurality of airflow direction control fans, each disposed on one side of the plurality of outlets, respectively, and wherein the controller is configured to change the direction of the air discharged from the plurality of outlets, respectively, by regulating a rotational speed of the plurality of airflow direction control fans.

10. A method for operating an air conditioner having an inlet and a plurality of outlets disposed around a circumference of the inlet, the method comprising: determining, via a controller, an activity level of an occupant for each of a plurality of areas of an indoor space based on a plurality of images corresponding to the indoor space and captured by a camera of the air conditioner; determining, via the controller, a direction of air discharged from the plurality of outlets based on the determined activity level; and according to the direction of the air which is determined for the plurality of outlets, controlling an airflow direction controller configured to control an airflow direction of air flowing through the plurality of outlets, wherein the determining of the activity level of the occupant comprises calculating, via a controller, a motion vector for the plurality of pixels constituting the occupant and included in the plurality of images, and wherein the activity level of the occupant is determined based on a value obtained by dividing a sum of magnitudes of motion vectors for the plurality of pixels, which are calculated for the plurality of images, by a number of the pixels.

11. The method of claim 10, wherein the controlling of the airflow direction controller comprises, among the plurality of areas, based on an activity level for a first area being greater than a predetermined level and an activity level for a second area being less than the predetermined level, controlling the airflow direction controller so that a direction of air discharged from a first outlet corresponding to the first area is different from a direction of air discharged from a second outlet corresponding to the second area.

12. The method of claim 11, wherein the controlling of the airflow direction controller comprises: during a cooling operation, controlling the airflow direction controller so that the direction of the air discharged from the first outlet is vertically below the direction of the air discharged from the second outlet; and during a heating operation, controlling the airflow direction controller so that the direction of the air discharged from the first outlet is vertically above the direction of the air discharged from the second outlet.

13. The method of claim 11, wherein the controlling of the airflow direction controller further comprises, among the plurality of areas, based on an activity level corresponding to a third area satisfying a criterion associated with the predetermined level, controlling the airflow direction controller so that a direction of air discharged from a third outlet corresponding to the third area changes continuously between the direction of the air discharged from the first outlet and the direction of the air discharged from the second outlet.

14. The method of claim 13, wherein a distance between the third area and the air conditioner is

longer than at least one of a distance between the first area and the air conditioner or a distance between the second area and the air conditioner.

15. The method of claim 10, further comprising: setting a target temperature based on a result of comparison between a predetermined reference and a sum of activity levels of the occupant which are determined for each of the plurality of areas; and controlling a temperature of an indoor space based on a difference between a current temperature of the indoor space, which is sensed by a temperature sensor that senses the temperature of the indoor space, and the target temperature.

16. The method of claim 10, wherein the determining of the activity level of the occupant comprises: calculating a motion vector using an optical flow method.

17. The method of claim 10, wherein the airflow direction controller comprises a plurality of vanes disposed at the plurality of outlets, respectively, and wherein the controlling of the airflow direction controller comprises changing the direction of the air discharged from the plurality of outlets by adjusting an arrangement of each of the vanes.

18. The method of claim 10, wherein the airflow direction controller comprises a plurality of airflow direction control fans, each disposed at one side of the plurality of outlets, respectively, wherein the controlling of the airflow direction controller comprises changing the direction of the air discharged from the plurality of outlets by regulating a rotational speed of the plurality of airflow direction control fans.
